

## Skills Review

### Exercise No. 5

Name: Aaron Ludwig A. Altar

Grade:

Course/Section: CPE4B

Date: Sept. 12, 2026

#### Machine Problem:

Create a program that will display the output format shown below. Save the program using Exer5.asm as the filename.

**zAyBxCwDu..... aZ**

#### Program:

```
; AARON LUDWIG A. ALTAR
; CPE4B

.model small
.stack
.code

start:
    mov ah, 02h
    mov cx, 26      ; 26 letters, A to Z

    mov bl, 41h     ; bl = 'A'
    mov dl, 7ah     ; dl = 'z'
    mov dh, dl      ; dh = 'z'

back:
    int 21h         ; display uppercase (dl)
    mov dl, bl      ; load lowercase
    int 21h         ; display lowercase

    inc bl          ; next uppercase letter
    dec dh          ; next lowercase letter

    mov dl, dh
    loop back

    mov dl, 0dh
    int 21h
    mov dl, 0ah
```

```
        int 21h

        mov ah, 4ch
        int 21h
end start
```

#### Output:

```
D:\>IF NOT ERRORLEVEL 1 TLINK D:\test
Turbo Link  Version 7.1.30.1. Copyright (c) 1987, 1996 Borland International
D:\>IF NOT ERRORLEVEL 1 D:\test
zAyBxCwDvEuFtGsHrIqJpKoLnMmNlOkPjQiRhSgTfUeVdWcXbYaZ
D:\>
```